CadetX Heavy Supplier, Inventory, and Warehouse Analytics: Final Project Report

Program: CadetX Data Engineering & Analytics Program

Phase: 3 — Advanced Analytics & Business Intelligence

Teams:

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1. **Executive Summary**

Over the course of this 12-week intensive program, our team engineered an end-to-end data pipeline, predictive demand forecasting model, and business intelligence (BI) dashboard suite tailored for heavy supply chain operations. This project successfully addressed critical supply chain bottlenecks, inventory distribution inefficiencies, and customer concentration vulnerabilities across multi-warehouse logistics networks, establishing a robust, scalable framework to optimize commercial performance and operational readiness.

2. **Core Architectural & Analytical Modules**

Data Ingestion & Cleaning (Weeks 1–4): Standardized raw transactional logs, supplier records, and warehouse inventory databases, executing rigorous data quality checks, handling missing values, and establishing a unified relational schema.

Exploratory Data Analysis & RFM Segmentation (Weeks 5–8): Segmented buyer behavior using Recency, Frequency, and Monetary (RFM) modeling, identifying high-value customer deciles, core champions, and revenue concentration drivers.

Predictive Forecasting & BI Dashboards (Weeks 9–11): Deployed time-series predictive algorithms forecasting an 8% Q1 growth trajectory ($520,000 revenue target) and built interactive executive BI dashboards.

Consolidation & Strategic Delivery (Week 12): Finalized repository governance, executive reporting, and strategic operational roadmaps to communicate business value effectively.

3. Comprehensive Key Findings & Analytical Insights

Commercial & Sales Performance: Total revenue reached $1,842,500 across 12,450 orders and 48,200 total units sold, maintaining a robust Average Order Value (AOV) of $147.99 and a 64.2% repeat customer rate. The top 20% of catalog SKUs account for 74% of total revenue, matching a strict Pareto distribution.

Customer Concentration & RFM Segmentation: Customer revenue is highly concentrated; the top decile (10%) of customer accounts generates 58% of overall gross revenue. RFM cohorts indicate that “Champions” and “Loyal Customers” form the core engine of repeat purchase stability.

Warehouse Operational Bottlenecks: The Central Warehouse handles 45% of total order processing volume but exhibits a disproportionate 4.2% fulfillment delay rate compared to regional hub averages (ranging between 1.5% and 2.1%).

Predictive Forecasting & Demand Analysis: Time-series demand modeling projects a stable Q1 revenue expansion of approximately 8%, targeting $520,000 as regional inventory rebalancing takes effect.

4. Strategic Business Recommendations

Inventory Decentralization: Shift 15% of buffer stock for high-velocity SKUs out of the Central Warehouse and into under-utilized regional hubs to alleviate capacity stress and lower fulfillment delay rates below 2%.

Tiered Customer SLAs: Implement structured Service Level Agreements (SLAs) and dedicated account management for the top 10% customer decile to protect key revenue drivers and mitigate churn risk.

SKU Rationalization: Review low-performing catalog items residing outside the top Pareto thresholds to optimize warehouse storage costs and free up working capital.

5. Project Limitations & Future Improvements

Limitations: The current analytical model relies heavily on historical static logs and lacks real-time Internet of Things (IoT) sensor feeds from freight carriers.

Future Improvements: Upgrade the ingestion layer to a real-time streaming pipeline (using Kafka or cloud event hubs) and integrate machine learning models capable of dynamic, automated warehouse routing based on live traffic and weather conditions.